

Process scheduling examples:

Q1: Find the average waiting time and average turnaround time for executing the following processes using FCFS (first-come first-service) scheduling?

Process	Burst time
P0	7
P1	5
P2	2
P3	9

Answer:

The first step of the solution is founding the Gantt chart.

Gantt chart:

P0	P1	P2	P3
0	7	12	14
			23

The waiting time of P0 = 0

The waiting time of P1 = 7

The waiting time of P2 = 12

The waiting time of P3 = 14

The average waiting time = $(0 + 7 + 12 + 14) / 4 = 33/4 = 8.25$

The turnaround time of p0 = 7

The turnaround time of p1 = 12

The turnaround time of p2 = 14

The turnaround time of p3 = 23

The average turnaround time = $(7+12+14+23)/4 = 56/4 = 14$

Q2: Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process using (1) Preemptive short-job first (2) Non-preemptive short-job first?

Process	P1	P2	P3	P4	P5
Burst time	5	13	8	4	10
Arrival time	2	3	0	5	1

Answer

(1) Using preemptive short-job first

Gantt chart

P3	P1	P4	P3	P5	P2	
0	2	7	11	17	27	40

$$\text{W.T. of p1} = 2 - 2 = 0$$

$$\text{T.A.T. of P1} = 7 - 2 = 5$$

$$\text{W.T. of p2} = 27 - 3 = 24$$

$$\text{T.A.T. of P2} = 40 - 3 = 37$$

$$\text{W.T. of p3} = 0 + (11 - 2) = 9$$

$$\text{T.A.T. of P3} = 17 - 0 = 17$$

$$\text{W.T. of p4} = 7 - 5 = 2$$

$$\text{T.A.T. of P4} = 11 - 5 = 6$$

$$\text{W.T. of p5} = 17 - 1 = 16$$

$$\text{T.A.T. of P5} = 27 - 1 = 26$$

$$\text{A.W.T.} = (0 + 24 + 9 + 2 + 16) / 5 = 51 / 5 = 10.2$$

$$\text{A.T.A.T} = (5 + 37 + 17 + 6 + 26) / 5 = 91 / 5 = 18.2$$

2) Using non-preemptive short-job first

Gantt chart

P3	P4	P1	P5	P2	
0	8	12	17	27	40

$$\text{W.T. of } p_1 = 12 - 2 = 10$$

$$\text{T.A.T. of } P_1 = 17 - 2 = 15$$

$$\text{W.T. of } p_2 = 27 - 3 = 24$$

$$\text{T.A.T. of } P_2 = 40 - 3 = 37$$

$$\text{W.T. of } p_3 = 0$$

$$\text{T.A.T. of } P_3 = 8 - 0 = 8$$

$$\text{W.T. of } p_4 = 8 - 5 = 3$$

$$\text{T.A.T. of } P_4 = 12 - 5 = 7$$

$$\text{W.T. of } p_5 = 17 - 1 = 16$$

$$\text{T.A.T. of } P_5 = 27 - 1 = 26$$

$$\text{A.W.T.} = (10 + 24 + 0 + 3 + 16) / 5 = 53 / 5 = 10.6$$

$$\text{A.T.A.T} = (15 + 37 + 8 + 7 + 26) / 5 = 93 / 5 = 18.6$$

Q3: Find the average waiting time and turnaround time for executing the following process using priority scheduling algorithm?

Process	P1	P2	P3	P4	P5
Burst time	5	13	8	6	12
Priority	1	3	0	4	2

Answer

Gantt chart

P3	P1	P5	P2	P4	
0	8	13	25	38	44

W.T. of p1 = 8

T.A.T. of P1=13

W.T. of p2 = 25

T.A.T. of P2= 38

W.T. of p3 = 0

T.A.T. of P3= 8

W.T. of p4 = 38

T.A.T. of P4= 44

W.T. of p5 = 13

T.A.T. of P5= 25

$$A.W.T. = (8+25+0+38+13)/5 = 84/5 = 16.8$$

$$A.T.A.T = (13+38+8+44+25)/5 = 128/5 = 25.6$$

Q4: Find the average waiting time (A.W.T.) and the average turnaround time (A.T.A.T.) for executing the following processes using round-robin algorithm, where time quantum is 5?

Process	P1	P2	P3	P4	P5
Burst time	11	4	14	9	21
Arrival time	5	0	0	1	2

Answer

$$\begin{aligned}
 P2 &= \cancel{14} \quad 0 \\
 P3 &= \cancel{14} \quad \cancel{4} \quad 0 \\
 P4 &= \cancel{9} \quad \cancel{4} \quad 0 \\
 P5 &= \cancel{21} \quad \cancel{14} \quad \cancel{11} \quad \cancel{6} \quad \cancel{4} \quad 0 \\
 P1 &= \cancel{11} \quad \cancel{6} \quad \cancel{4} \quad 0
 \end{aligned}$$

Gantt chart

P2	P3	P4	P5	P1	P3	P4	P5	P1	P3	P5	P1	P5	P5
0	4	9	14	19	24	29	33	38	43	47	52	53	58 59

$$\text{W.T. of P1} = (19-5) + (38-24) + (52-43) = 14 + 14 + 9 = 37$$

$$\text{W.T. of P2} = 0$$

$$\text{W.T. of P3} = (4-0) + (24-9) + (43-29) = 4 + 15 + 14 = 33$$

$$\text{W.T. of P4} = (9-1) + (29-14) = 8 + 15 = 23$$

$$\text{W.T. of P5} = (14-2) + (33-19) + (47-38) + (53-52) = 12 + 14 + 9 + 1 = 36$$

$$\text{A.W.T.} = (37 + 0 + 33 + 23 + 36) / 5 = 129 / 5 = 25.8$$

$$\text{T.A.T of P1} = (53-5) = 48$$

$$\text{T.A.T of P2} = (4-0) = 4$$

$$\text{T.A.T of P3} = (47-0) = 47$$

$$\text{T.A.T of P4} = (33-1) = 32$$

$$\text{T.A.T of P5} = (59-2) = 57$$

$$\text{A.T.A.T.} = (48 + 4 + 47 + 32 + 57) / 5 = 188 / 5 = 37.6$$

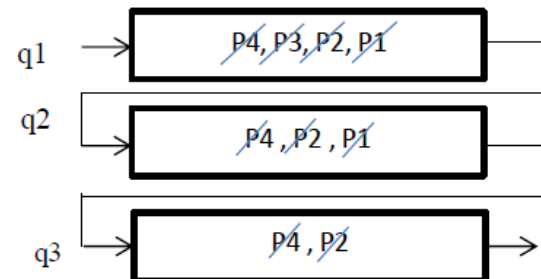
Q25: Consider a multilevel feedback queue scheduling (MLFBQ) with three queues q1, q2, and q3. q1 and q2 use round-robin algorithm with time quantum (TQ) = 5, and 4 respectively. q3 use first-come first-service algorithm. Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process?

Processes	P1	P2	P3	P4
Burst time	8	22	4	12

Answer

In MLFBQ scheduling algorithm, the process move between queues. If a process uses too much CPU time, it will be moved to a lower-priority queue.

	q1	q2	q3
P1:	8	3	-
P2:	22	17	13
P3:	4	-	-
P4:	12	7	3



Gantt chart

P1	P2	P3	P4	P1	P2	P4	P2	P4	
0	5	10	14	19	22	26	30	43	46

$$\text{W.T. of p1} = 0 + (19-5) = 14$$

$$\text{T.A.T. of P1} = 22$$

$$\text{W.T. of p2} = 5 + (22-10) + (30-26) = 5+12+4 = 21$$

$$\text{T.A.T. of P2} = 43$$

$$\text{W.T. of p3} = 10$$

$$\text{T.A.T. of P3} = 14$$

$$\text{W.T. of p4} = 14 + (26-19) + (43-30) = 14+7+13 = 34$$

$$\text{T.A.T. of P4} = 46$$

$$\text{A.W.T.} = (14 + 21 + 10 + 34)/4 = 79/4 = 19.75$$

$$\text{A.T.A.T} = (22 + 43 + 14 + 46)/4 = 125/4 = 31.25$$